JAEEON LEE

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EDUCATION & RESEARCH

Harvard University, Department of Molecular & Cellular Biology, MA — 10/2022~Present
Postdoctoral Research Fellow
Advisors: Naoshige Uchida, Samuel Gershman

10/2021 ~ 9/2022

Columbia University, Zuckerman Institute, NY
Postdoctoral Research Fellow
Advisor: Rui Costa

Harvard University, Department of Neurobiology, MA — 09/2015 ~ 9/2021
Ph. D. in Neuroscience,
Advisor: Bernardo Sabatini

KAIST, Department of Bio and Brain Engineering, Korea — 09/2010 ~ 02/2015
B. S. in Bio and Brain Engineering (*Magnum cum laude*)

HONORS & AWARDS

K99/R00 Pathway to Independence award	2026
Astera Essay Competition Winner	2026
Harold M. Weintraub Graduate Student Award	2022
Harvard Brain Initiative Travel Award (visit to Branco's lab)	2019
Iljou foundation Scholarship for Ph.D (5 years)	2015-2022

CONFERENCES & TALKS

Basal Ganglia GRC 2026, Poster Presentation — 2026
Emergence of rapid value inference via meta-RL
Cosyne Meeting 2026, Poster Presentation — 2026
Emergence of rapid value inference via meta-RL
Sfn meeting San Diego, Poster presentation — 2025
Emergence of rapid value inference via meta-RL
VIDA Dopamine conference 2021, Poster, Virtual Conference — 2021
Striatal indirect pathway mediates exploration via modulation of collicular dynamics
Brain Initiative Meeting 2021, Poster, Virtual Conference — 2021
Striatal indirect pathway mediates exploration via modulation of collicular dynamics
Cosyne 2021, Poster, Virtual Conference — 2021
Striatal indirect pathway mediates exploration via modulation of collicular dynamics
Neurobiology Departmental Seminar, Talk, Harvard Medical School, MA, US — 2019

Investigating the computation of the striatal indirect pathway

Tiago Branco's lab, Talk, Sainsbury Wellcome Centre, London, United Kingdom 2019

Investigating the computation of the striatal indirect pathway

Society for Neuroscience 2018, Poster, San Diego, US 2018

Mapping the basal ganglia topography

PUBLICATIONS

Lee, J*, Hennig, J., Frelih, V., Gershman, Sam., Uchida, N. *Experience shifts value updating from synaptic plasticity to neural dynamics.* (2026, *Biorxiv) (In Review*)

Lee, J.*, Sabatini, B.L. From avoidance to new action: the multifaceted role of the striatal indirect pathway. *Nat. Rev. Neurosci.* 26, 438–449 (2025).

Lee, J.*, Sabatini, B. Striatal indirect pathway mediates exploration via collicular competition. *Nature.* (2021).

Lee, J.*, Wang, W., Sabatini, B. Anatomically segregated basal ganglia pathways allow parallel behavioral modulation. *Nat. Neurosci.* 1–11 (2020).

Pisano, F.*, Pisanello, M.*, Lee, S.J. **Lee, J.**, *et al,.* Depth-resolved fiber photometry with a single tapered optical fiber implant. *Nat Methods* 16**,** 1185–1192 (2019).

TEACHING

Black Undergraduate Mentorship Program (BUMP) mentor 2/2022

Teaching assistant for Matlab Bootcamp (assisted programming in Matlab) 6/2019

Teaching assistant for Thinking about Data (assisted programming in Matlab/statistical analysis) 9/2018

REFERENCES

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